

Heart Transplantation

Executive Summary

Heart transplantation remains the definitive therapy for carefully selected patients with end-stage heart failure whose prognosis and quality of life remain poor despite contemporary guideline-directed medical therapy, device therapy, and specialist advanced heart failure care. Contemporary practice is increasingly shaped by stricter candidate optimisation, broader but more nuanced donor acceptance, routine use of mechanical circulatory support as a bridge to transplant or to candidacy, and expanding use of donation after circulatory death. Registry data indicate that heart transplant activity has risen again in recent years: within the relaunched ISHLT registry, an average of roughly 2500–3000 heart transplants were performed annually among contributing partners, rising to 5,887 reported procedures in 2023; in the United States, heart transplants reached a record 4,636 in 2024. Recipient and donor profiles have also evolved, with older recipients, modestly older donors, and greater use of DCD organs and pre-transplant mechanical support than in earlier eras. [1]

Outcomes are now strong in experienced programmes, but they remain constrained by early graft dysfunction and by long-term complications of chronic immunosuppression. In the current U.S. national report, adult post-transplant survival was 91.2% at 1 year and 80.1% at 5 years; in the contemporary UK adult report, national 1-year survival was 89.0% and 5-year survival 74.8%, while UK paediatric survival was 92.9% at 1 year and 85.7% at 5 years. Long-term adult median survival in international registry analyses remains approximately 12 years overall and longer among recipients surviving the first post-transplant year. Late morbidity is dominated by cardiac allograft vasculopathy, chronic kidney disease, infection, metabolic disease, and malignancy. [2]

The central strategic challenge in the field is no longer whether heart transplantation works, but how to extend its benefits safely and more equitably. The most credible near-term advances are broader donor utilisation, especially DCD heart recovery, more personalised immunosuppression, non-invasive rejection surveillance, and better long-term prevention of allograft vasculopathy and malignancy. Longer-horizon innovations include clinical cardiac xenotransplantation and immune tolerance strategies using mixed chimerism, co-stimulation blockade, and regulatory cell therapies. [3]

Title Page and Abstract

Manuscript title: Contemporary Heart Transplantation: Epidemiology, Selection, Surgical Practice, Immunobiology, Outcomes and Future Directions

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Structured Abstract

Background: Heart transplantation is the reference standard for selected patients with advanced heart failure, but its use is constrained by donor scarcity, increasing recipient complexity, and the burden of long-term allograft surveillance and immunosuppression. Contemporary guidance from ISHLT, AHA and ESC has materially changed candidate evaluation, donor assessment, perioperative management and post-transplant monitoring.

Methods: A structured narrative review was undertaken using major society guidelines, official registry reports and peer-reviewed literature prioritising primary studies, consensus documents and reviews published mainly between 2015 and July 2026, with older landmark studies retained where necessary for key concepts such as primary graft dysfunction and rejection classification.

Results: Current evidence supports transplantation in highly selected patients with refractory stage D heart failure after multidisciplinary assessment of haemodynamics, end-organ function, frailty, psychosocial readiness and competing comorbidity. Donor selection has broadened through better size matching, coronary screening in older donors, and use of DCD recovery pathways. Bicaval orthotopic transplantation is generally preferred over biatrial implantation because of more favourable early and late outcomes, while heterotopic transplantation is now rare and reserved for exceptional anatomical or pulmonary vascular scenarios. Standard maintenance immunosuppression remains calcineurin inhibitor-based, usually tacrolimus with mycophenolate and corticosteroids, with selective use of induction therapy and mTOR-based renal-sparing or vasculopathy-targeted strategies. Endomyocardial biopsy remains the cornerstone of rejection diagnosis, increasingly complemented by gene-expression profiling and donor-derived cell-free DNA in selected patients. Contemporary survival is excellent at 1 year in North America and Western Europe, but late complications including cardiac allograft vasculopathy, chronic kidney disease, infection and malignancy continue to limit durability.

Conclusions: The field is moving towards individualised donor-recipient matching, risk-adapted immunosuppression, expanded donor pools, and translational programmes in xenotransplantation and tolerance induction; however, equitable access, ethical governance, and prevention of late graft failure remain decisive priorities. [4]

Keywords: heart transplantation; advanced heart failure; donor selection; immunosuppression; rejection; cardiac allograft vasculopathy; xenotransplantation

Introduction and Methods

Introduction: Heart transplantation occupies a distinct place in cardiovascular therapeutics: it is not first-line management for heart failure, but it remains the best established life-extending and quality-of-life-restoring intervention for appropriately selected patients with refractory advanced disease. The 2024 ISHLT candidate guideline, the 2023 ISHLT recipient-care guideline, the 2021 ESC heart failure guideline with 2023 focused update, and the AHA statement on timely referral for advanced heart failure collectively frame current practice: optimise medical and device therapy early, refer before irreversible end-organ injury occurs, and use transplant only when the anticipated survival and functional gain clearly justify scarce-organ use. [5]

The epidemiological context is sobering. Heart transplantation is technically mature, but globally limited by donor availability and infrastructure. A recent review estimated that more than 7,000 heart transplants were performed worldwide in 2020 and that about 90% were concentrated in the United States and Western Europe. In the relaunched ISHLT international registry, 5,887 transplants were reported in 2023, and the adult median recipient age across 1992–2024 was 55 years, with a median donor age of 32 years; 2.4% of adult recipients received DCD hearts. This concentration of activity means that “heart transplant outcomes” are heavily shaped by geography, allocation systems, donor utilisation, and access to mechanical circulatory support. [6]

Methods: This manuscript is a structured narrative review designed for journal-style submission. Sources were prioritised in the following hierarchy: major international guidelines and consensus documents; official national and international registry reports; peer-reviewed primary studies; and contemporary review articles. Searches emphasised publications from 2015 onward, while older landmark evidence was retained selectively for concepts that still structure present-day care, including pulmonary vascular resistance thresholds, AMR nomenclature, and surgical technique comparisons. Evidence was synthesised qualitatively rather than meta-analytically, and where current sources reported non-identical survival metrics, ranges are stated instead of a single pooled estimate. [7]

Results and Discussion

Epidemiology and indications: The modern indication for heart transplantation is refractory advanced heart failure, usually corresponding to stage D disease with severe limitation, repeated decompensation, dependence on inotropes or temporary support, refractory ventricular arrhythmias, or deteriorating end-organ function despite comprehensive therapy. The AHA referral statement stresses that referral should occur before irreversible renal, hepatic, pulmonary or frailty-related decline, while the ESC guideline embeds transplantation within the broader advanced heart failure pathway alongside durable mechanical support and palliative care. Transplantation may also be indicated for selected patients with congenital heart disease, restrictive or hypertrophic cardiomyopathy, severe graft failure after prior transplant, or highly selected inherited arrhythmia syndromes with otherwise unacceptable prognosis. [8]

Donor selection: Donor assessment has shifted from rigid “ideal donor” rules to risk contextualisation. The ISHLT donor guideline recommends preferential use of donor hearts under 45 years of age; donors aged 45 years and older can still be acceptable after screening for significant coronary artery disease and when cold ischaemic time can be kept short. Predicted heart mass has emerged as the preferred size-matching metric, outperforming height or weight alone in adult transplantation. Several donor characteristics once treated as near-automatic exclusions are now handled more selectively, particularly in an era of donor shortage, organ perfusion technology and improved perioperative support. DCD hearts have become the clearest example of successful donor-pool expansion: in the randomised trial comparing DCD with donation after brain death, risk-adjusted 6-month survival was non-inferior at 94% versus 90%, and international contemporary observational data suggest that thoraco-abdominal normothermic regional perfusion is associated with 1-year survival around 91%, comparable to or slightly better than direct procurement and perfusion. [9]

Recipient selection: Recipient evaluation is now explicitly holistic. The 2024 ISHLT candidate guideline emphasises cardiac severity, reversibility of comorbidity, psychosocial reliability and waiting-list optimisation rather than a simplistic checklist. Several threshold-type findings remain highly influential. In adults with BMI at or above 35 kg/m², weight loss to a BMI below 35 kg/m² is considered reasonable before listing. In candidates with pulmonary artery systolic pressure at or above 50 mmHg, together with either transpulmonary gradient at or above 15 mmHg or pulmonary vascular resistance at or above 3 Wood units, stepwise haemodynamic assessment is recommended because fixed pulmonary vascular disease places the donor right ventricle at substantial risk. Established GFR below 30 mL/min/1.73 m² is a major warning sign and should prompt evaluation for reversibility and, where appropriate, consideration of combined heart-kidney transplantation. Frailty matters independently: pre-transplant frailty within six months of transplantation has been associated with higher mortality and longer hospitalisation, and current ISHLT guidance strongly supports its routine assessment. Active uncontrolled infection, non-adherence, uncontrolled substance use, severe

progressive extracardiac disease, and absent psychosocial support remain major barriers to isolated heart transplantation. [10]

Surgical techniques: Orthotopic transplantation is the standard operation. Within orthotopic transplantation, bicaval implantation has largely supplanted the classic biatrial method because meta-analysis shows superior early and late outcomes, less tricuspid regurgitation, less atrial distortion and fewer pacemaker requirements. The biatrial technique remains important historically and may still be used in selected redo or technically complex cases, but it is no longer the preferred routine approach in most programmes. Heterotopic transplantation, in which the donor heart is implanted in parallel with the native heart, is now rare and mostly reserved for exceptional situations such as severe pulmonary hypertension, marked size mismatch, or when temporary biventricular assistance from the native heart may be advantageous. Its rarity reflects progress in candidate optimisation, pulmonary vascular management and mechanical circulatory support rather than a lack of conceptual utility. [11]

Perioperative management: Early post-transplant care begins with meticulous haemodynamic and rhythm monitoring. ISHLT recommends continuous ECG, postoperative 12-lead ECG, invasive arterial pressure monitoring, direct right-sided filling pressure measurement, and left-sided pressure or wedge pressure monitoring in appropriate recipients. Primary graft dysfunction remains the dominant early cause of severe morbidity and early death. The current ISHLT graft dysfunction update aims to standardise prevention, diagnosis and management, while the recipient-care guideline recommends very early mechanical circulatory support when separation from bypass fails or when worsening haemodynamic instability suggests graft failure. Renal protection is built into contemporary protocols: if significant perioperative renal dysfunction is present, delayed initiation of calcineurin inhibition is reasonable. Metabolic and infectious prophylaxis are now similarly protocolised; one cited ISHLT target is perioperative blood glucose control in the range of 140–180 mg/dL. CMV prophylaxis is recommended early after transplant, especially in D+/R– mismatches, and Pneumocystis prophylaxis with trimethoprim-sulfamethoxazole is recommended for at least 6–12 months. [12]

Maintenance immunosuppression: The standard maintenance platform remains triple therapy with tacrolimus, mycophenolate and corticosteroids, although steroid minimisation is common after the early months in low-risk recipients. Tacrolimus trough targets in the ISHLT guideline are generally 10–15 ng/mL in days 0–60, 8–12 ng/mL during the next 3–6 months, and 5–10 ng/mL beyond 6 months in stable recipients using azathioprine or mycophenolic acid as adjuncts. mTOR-based regimens are not routine first-line therapy for every patient, but they are strategically important. When combined with calcineurin inhibitors, reduced CNI exposure is recommended; tacrolimus targets may be lowered by about 40–50%, and everolimus trough targets of 3–8 ng/mL are considered reasonable. These strategies are attractive in recipients with renal dysfunction, recurrent rejection, or established or high-risk cardiac allograft vasculopathy. A 2025

review concluded that immunosuppression should now be framed not as a fixed regimen but as a dynamic risk-adapted programme balancing rejection, infection, nephrotoxicity, malignancy and vasculopathy. Induction therapy remains controversial: contemporary evidence reviews commissioned for guideline work found persistent uncertainty about the net benefit of universal induction, so practice remains selective, usually favouring higher immunological risk or renal-sparing scenarios. [13]

Rejection diagnosis and management: Endomyocardial biopsy remains the diagnostic cornerstone for acute rejection, particularly in the first months after transplantation and in higher-risk patients. Yet surveillance is clearly becoming more layered. ISHLT supports the use of gene-expression profiling in low-risk adult recipients between 2 months and 5 years after transplant to reduce biopsy frequency, and donor-derived cell-free DNA has been validated prospectively as a non-invasive marker of graft injury. After the first year, continued surveillance using some combination of non-invasive methods, GEP or biopsy is considered reasonable in recipients at higher risk for late acute rejection, including those with prior rejection or donor-specific antibodies. Treatment remains phenotype-based. For asymptomatic severe cellular rejection (3R), high-dose intravenous corticosteroids are recommended; asymptomatic 2R rejection is typically treated with intravenous or oral corticosteroids, while recurrent or steroid-resistant cellular rejection should prompt cytolytic therapy such as antithymocyte globulin and reassessment of the baseline regimen. AMR is more heterogeneous. Recent ISHLT consensus work emphasises antibody burden, antibody characteristics, biopsy phenotype, graft dysfunction and overall patient risk rather than a single binary treatment rule. Contemporary AMR management typically combines augmentation of baseline immunosuppression with corticosteroids and, in more clinically significant cases, plasmapheresis, intravenous immune globulin, rituximab and selected targeted therapies. [14]

Complications: Early complications are dominated by primary graft dysfunction, vasoplegia, bleeding, renal failure and severe infection. Later complications reflect both chronic alloimmunity and chronic immunosuppression. Cardiac allograft vasculopathy is the signature long-term graft disease; its prevention depends partly on risk factor control and partly on immunobiology. ISHLT recommends annual or biannual coronary angiography in adults, with baseline angiography and often IVUS at 4–6 weeks and again at 1 year in order to exclude donor-derived coronary disease, detect rapidly progressive CAV and improve prognostic stratification. Statin therapy reduces CAV and remains part of standard preventive care, while early conversion to mTOR-based regimens may be considered for CAV prevention or treatment. Malignancy remains a major late burden: adult registry data cited in the 2023 ISHLT recipient guideline report cumulative prevalence of all malignancies of 16% in 5-year survivors and 28% in 10-year survivors, with skin cancer the most common. Chronic kidney disease is likewise frequent because of pre-existing haemodynamic injury, perioperative AKI and chronic calcineurin exposure, and it increasingly drives decisions on renal-sparing protocols and combined organ transplantation. [15]

Outcomes and quality of life: Survival continues to improve, but not uniformly. Older international registry work reported median adult survival of 12.5 years for transplants performed in the 2002–2009 era and 14.8 years among 1-year survivors; ISHLT fast facts report a median of 11.9 years using more recent summary reporting conventions. In the United States, current adult 1- and 5-year survival are 91.2% and 80.1%. In the UK, adult 1-year survival is approximately 89% and 5-year survival 74.8%, while adult median survival stratified by era improved from 11.9 years in 1995/96–1999/00 to 14.5 years in 2005/06–2009/10. These figures underline an important methodological point: survival estimates differ according to jurisdiction, calendar era, case mix and whether the measure is patient survival, graft survival, or conditional survival. Quality-of-life gains are nonetheless substantial. Systematic reviews conclude that health-related quality of life improves markedly after transplantation, particularly in physical function and symptom burden, although long-term survivors may still report reduced physical QoL relative to population norms. Rehabilitation is now considered routine rather than optional; ISHLT recommends structured cardiac rehabilitation, and recent consensus work emphasises long-term prevention, psychosocial support, functional reconditioning and return-to-participation as core components of transplant care. [16]

Ethical and legal issues: Ethical governance in heart transplantation begins before procurement and continues beyond allocation. The WHO Guiding Principles and the Declaration of Istanbul anchor international expectations around informed consent, voluntary non-remunerated donation, prohibition of trafficking and commercialism, transparency, and equitable stewardship of organs. DCD heart transplantation has revived focused ethical debate because of the dead donor rule and the permissibility of reperfusion-based recovery techniques; contemporary analyses conclude that DCD can be ethically justified when the determination of death is robust, procurement does not cause death, and governance is transparent and consistent with local law. Allocation ethics are also evolving. In the United States, continuous distribution proposals specifically aim to reduce inequity for sensitised candidates by awarding additional consideration to patients whose biology or prior exposure narrows donor access. These developments show that modern heart transplantation is inseparable from policy design: technical success alone does not guarantee fairness. [17]

Future directions: The near future of heart transplantation will depend on using more hearts, using them better, and eventually creating new graft sources. Mechanical circulatory support already functions as a bridge to transplant and a bridge to candidacy by stabilising shock, restoring end-organ function and, in some patients, lowering pulmonary vascular resistance sufficiently to permit isolated heart transplantation. DCD adoption is likely to continue expanding, supported by trial-level evidence of non-inferior short-term survival and by rapid-recovery approaches that may simplify logistics. Clinical cardiac xenotransplantation is no longer speculative science fiction: in 2026 ISHLT issued a dedicated consensus statement setting an evidence-based framework for candidate selection, oversight and trial conduct in pig-to-human heart transplantation. In parallel, tolerance induction remains the most important immunological frontier.

Contemporary reviews highlight mixed chimerism, co-stimulation blockade and regulatory T-cell therapies as the leading strategies to reduce lifelong dependence on calcineurin-based immunosuppression, though these approaches remain investigational in routine cardiac practice. [18]

Discussion: The field’s major achievement is that heart transplantation has become simultaneously safer and more ambitious: older and more complex recipients are now transplanted using broader donor criteria with survival that would once have seemed exceptional. The field’s major unresolved problem is that the same progress which expanded access has made long-term management more complex. Broader donor utilisation can increase primary graft dysfunction risk; intensified immunosuppression can suppress rejection but worsen infection, malignancy and renal failure; renal-sparing strategies may improve kidney outcomes but destabilise rejection control; and allocation reforms can improve equity while changing incentives and case mix. For clinicians, this means that the contemporary “best” transplant strategy is seldom universal. It is programme-specific, patient-specific and phase-specific. For health systems, it means that transplant success increasingly depends on the entire pathway: timely referral, waiting-list optimisation, donor recovery, perioperative rescue capability, surveillance infrastructure, and long-term survivorship care. [19]

Tables and Figures

Table 1. Contemporary donor and recipient selection considerations in adult heart transplantation

Domain	Favourable or acceptable considerations	Major caution or relative contraindication
Donor age	Donors <45 years are generally preferred.	Donors ≥45 years require more rigorous coronary and structural screening; careful attention to ischaemic time.
Donor coronary status	No significant CAD; normal or acceptably recoverable ventricular function.	Significant CAD or unrecoverable severe LV dysfunction reduce acceptability.
Size matching	Predicted heart mass is preferred for adult size matching.	Undersizing by PHM is associated with worse outcomes.
Donor pathway	DBD standard; DCD increasingly acceptable	DCD requires validated recovery pathway and programme expertise.

Domain	Favourable or acceptable considerations	Major caution or relative contraindication
	in experienced programmes.	
Recipient obesity	BMI <35 kg/m ² generally preferred.	BMI ≥35 kg/m ² should prompt weight-loss optimisation before listing.
Recipient pulmonary haemodynamics	Acceptable pulmonary vascular load or demonstrable reversibility.	PASP ≥50 mmHg with TPG ≥15 mmHg or PVR ≥3 WU requires formal stepwise assessment; persistent fixed pulmonary vascular disease may preclude isolated HT.
Renal function	Preserved or reversible dysfunction.	Established GFR <30 mL/min/1.73 m ² should trigger detailed renal assessment and possible combined-organ consideration.
Frailty and psychosocial readiness	Demonstrated adherence, social support, rehabilitation potential.	Significant frailty, non-adherence, uncontrolled psychiatric or substance-use disorder, or absent support can preclude listing.
Infection and malignancy	Treated or resolved infection; appropriately risk-stratified past malignancy.	Active uncontrolled infection or active malignancy are major barriers to isolated HT.

Data sources: ISHLT donor guideline, 2024 candidate guideline snippets and contemporary selection reviews. Thresholds and wording should be interpreted in programme context, not as isolated absolutes. [20]

Table 2. Common immunosuppressive agents in heart transplantation

Agent or class	Main role	Practical use in current care	Important advantages	Important limitations
Basiliximab	Selective induction	Used selectively in higher immunological risk or renal-sparing scenarios	Less lymphocyte depletion than ATG; lower infusion complexity	Universal benefit remains uncertain
Rabbit	Depleting	Selective induction;	Potent T-cell	Infection,

Agent or class	Main role	Practical use in current care	Important advantages	Important limitations
antithymocyte globulin	induction; resistant cellular rejection	treatment for recurrent or steroid-resistant ACR	depletion	cytopenia, infusion reactions
Tacrolimus	Core calcineurin inhibitor	Usually first-line maintenance with mycophenolate $\pm$ steroids	Strong anti-rejection efficacy; standard of care	Nephrotoxicity, neurotoxicity, diabetes, drug interactions
Cyclosporine	Alternative calcineurin inhibitor	Less commonly first-line in current adult practice	Familiar pharmacology	More hypertension, cosmetic effects, nephrotoxicity
Mycophenolate mofetil / mycophenolic acid	Antimetabolite adjunct	Standard partner to tacrolimus in triple therapy	Reduces rejection without CNI exposure increase	Gastrointestinal toxicity, cytopenias, teratogenicity
Azathioprine	Older antimetabolite	Alternative when mycophenolate is not tolerated or contraindicated	Simpler historical regimen	Less favoured than mycophenolate in modern practice
Prednisone / corticosteroids	Induction bridge, maintenance taper, rejection treatment	Universal very early use; withdrawal/minimisation common in selected patients	Rapid anti-inflammatory effect	Diabetes, weight gain, osteoporosis, infection
Everolimus	mTOR inhibitor	Used with reduced CNI for renal-sparing and CAV-focused strategies; EVL trough often 3–8 ng/mL with CNI	Renal-sparing potential; useful for CAV prevention/treatment	Dyslipidaemia, mouth ulcers, impaired wound healing, oedema

Agent or class	Main role	Practical use in current care	Important advantages	Important limitations
Sirolimus	mTOR inhibitor	Similar use to everolimus in selected patients	Helpful in CAV- or malignancy-oriented strategies	Wound-healing concerns, oedema, dyslipidaemia
Belatacept	Costimulation blockade	Investigational or highly selected non-standard use	Potential CNI avoidance	Limited heart-specific evidence; EBV-related caution

Key ISHLT target examples: tacrolimus 10–15 ng/mL in days 0–60, 8–12 ng/mL in months 2–6, and 5–10 ng/mL after 6 months in stable recipients using MPA/AZA; with mTOR-based regimens, tacrolimus targets may be reduced and EVL troughs of 3–8 ng/mL are commonly used. [21]

Table 3. Illustrative survival statistics by era and region

Cohort	Metric reported	Survival
International adult HT registry, 2002–2009 era	Median survival	12.5 years
International adult HT registry, conditional on 1-year survival	Median survival	14.8 years
ISHLT summary reporting	Adult median survival	11.9 years
United States, OPTN/SRTR 2024 report	Adult 1-year survival	91.2%
United States, OPTN/SRTR 2024 report	Adult 5-year survival	80.1%
United Kingdom, adult first HT, transplants 2020–2024	1-year survival	89.0%
United Kingdom, adult first HT, transplants 2016–2020	5-year survival	74.8%
United Kingdom, adult eras 1995/96–1999/00	Median survival	11.9 years
United Kingdom, adult eras 2000/01–2004/05	Median survival	13.5 years
United Kingdom, adult eras 2005/06–2009/10	Median survival	14.5 years

Cohort	Metric reported	Survival
United Kingdom, paediatric first HT, transplants 2020–2024	1-year survival	92.9%
United Kingdom, paediatric first HT, transplants 2016–2020	5-year survival	85.7%
Brazil, cited contemporary national series	1-year survival	about 70.9%–71%
Brazil, cited contemporary national series	5-year survival	about 59.5%

Source note: survival metrics are not directly interchangeable across registries because some reports present patient survival, others graft survival, and some provide conditional or era-specific medians. For manuscript text, these values should therefore be interpreted as a comparative range rather than a single pooled estimate. [22]

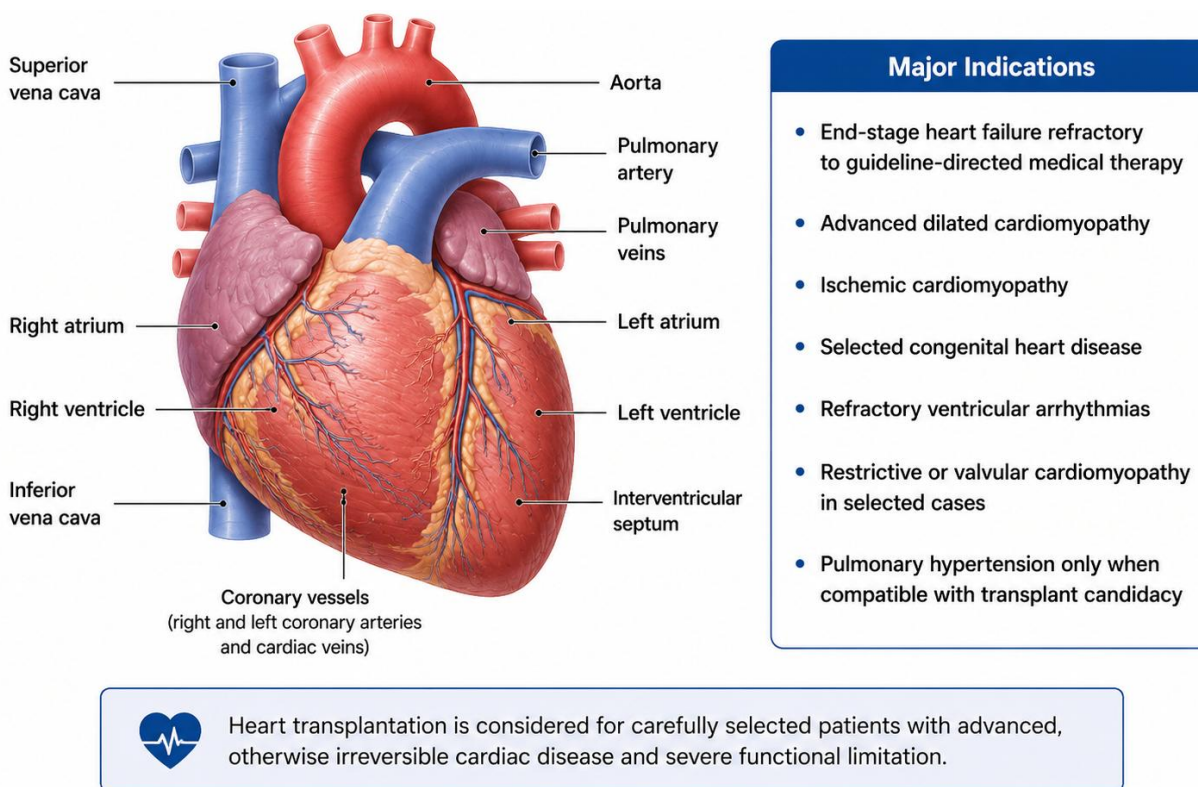
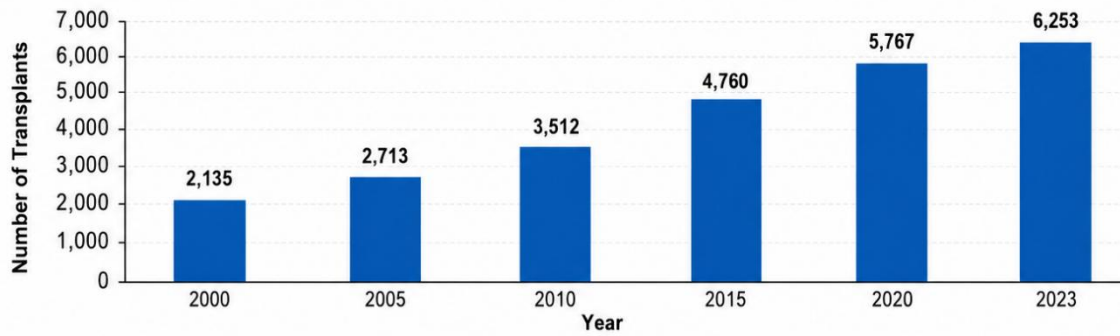
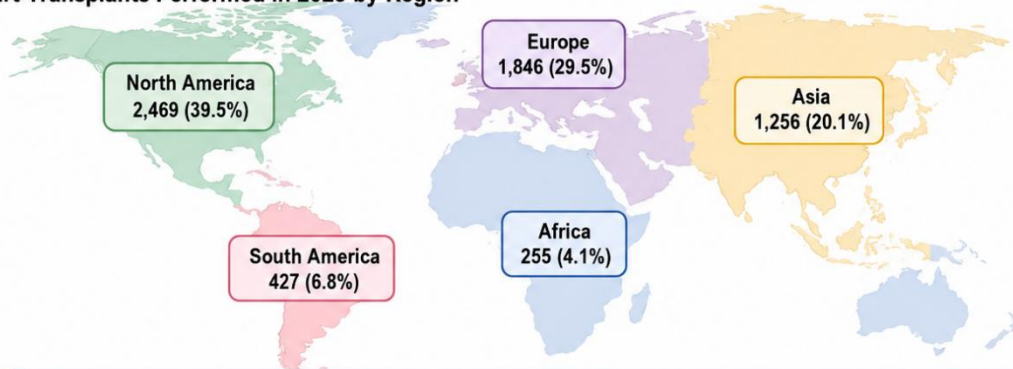


Figure 1. Anatomy of the Human Heart and Major Indications for Heart Transplantation

A. Annual Number of Heart Transplants Performed Worldwide

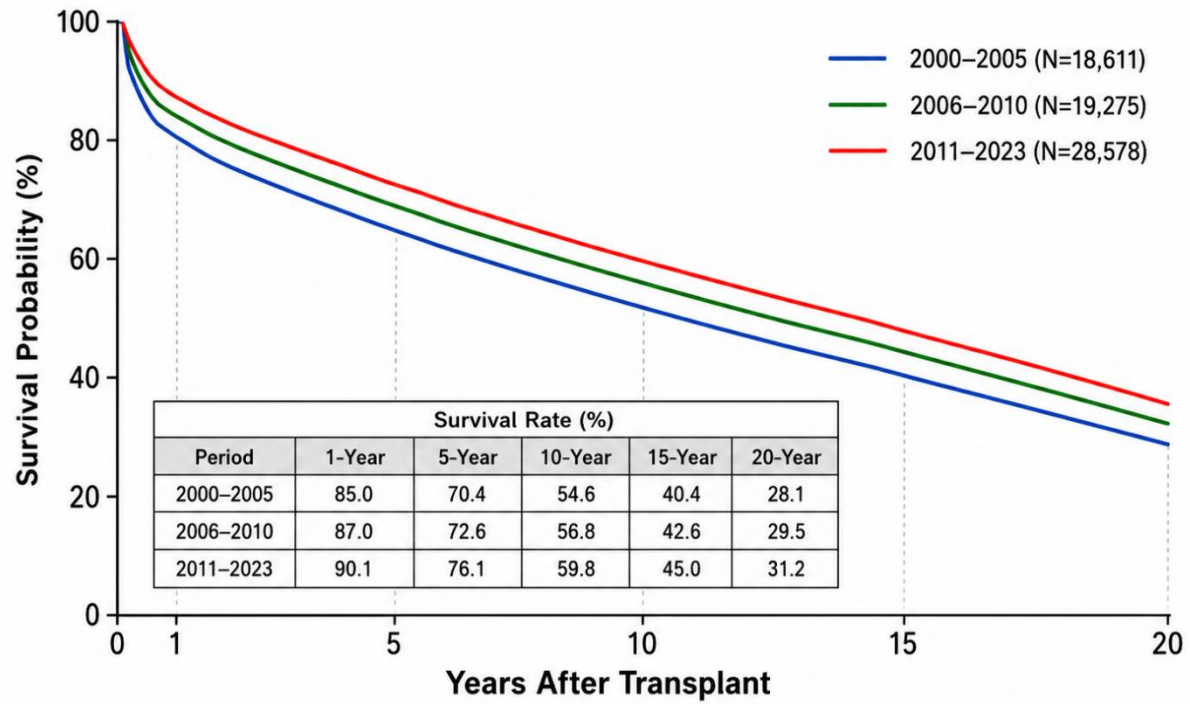


B. Heart Transplants Performed in 2023 by Region



Worldwide heart transplantation activity has increased substantially over the last two decades, with the largest volume reported in North America.

Figure 2. Global Trends in Heart Transplantation (2000-2023)



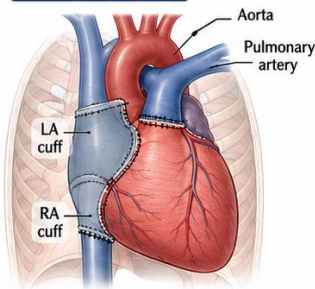
Post-transplant survival has improved over time owing to advances in donor selection, surgery, immunosuppression, and long-term follow-up.

Figure 3. Patient Survival After Heart Transplantation

1. Orthotopic Transplantation

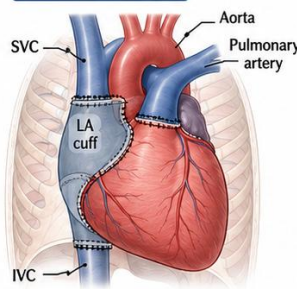
Recipient cardiectomy → donor heart implanted in native position

A. Biatrial Method



Connections (Anastomoses)			
donor LA cuff	↔	recipient LA cuff	
donor RA cuff	↔	recipient RA cuff	
aorta	↔	aorta	
pulmonary artery	↔	pulmonary artery	

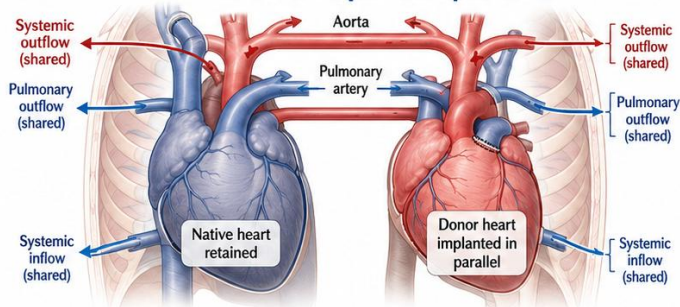
B. Bicaval Method



★ Preferred contemporary technique

Connections (Anastomoses)			
donor LA cuff	↔	recipient LA cuff	
donor SVC	↔	recipient SVC	
donor IVC	↔	recipient IVC	
aorta	↔	aorta	
pulmonary artery	↔	pulmonary artery	

2. Heterotopic Transplantation



Niche / exceptional use

- Native heart retained
- Donor heart implanted in parallel
- Shared systemic and pulmonary inflow/outflow support
- Reserved for exceptional cases such as severe pulmonary hypertension, size mismatch, or selected anatomically complex scenarios

Legend / Key

	Recipient anatomy (vena cava, atrial cuffs, other structures)
	Donor heart
	Oxygenated blood (arterial)
	Deoxygenated blood (venous)
	Anastomosis (suture line)

Caption: Bicaval orthotopic transplantation is generally preferred over the older biatrial technique because of superior preservation of atrial geometry, less tricuspid regurgitation and improved early and late outcomes. Heterotopic transplantation persists as a niche option rather than routine practice.

Figure 4. Described surgical schematic of orthotopic and heterotopic heart transplantation

Bicaval orthotopic transplantation is generally preferred over the older biatrial technique because of superior preservation of atrial geometry, less tricuspid regurgitation and improved early and late outcomes. Heterotopic transplantation persists as a niche option rather than routine practice. [23]

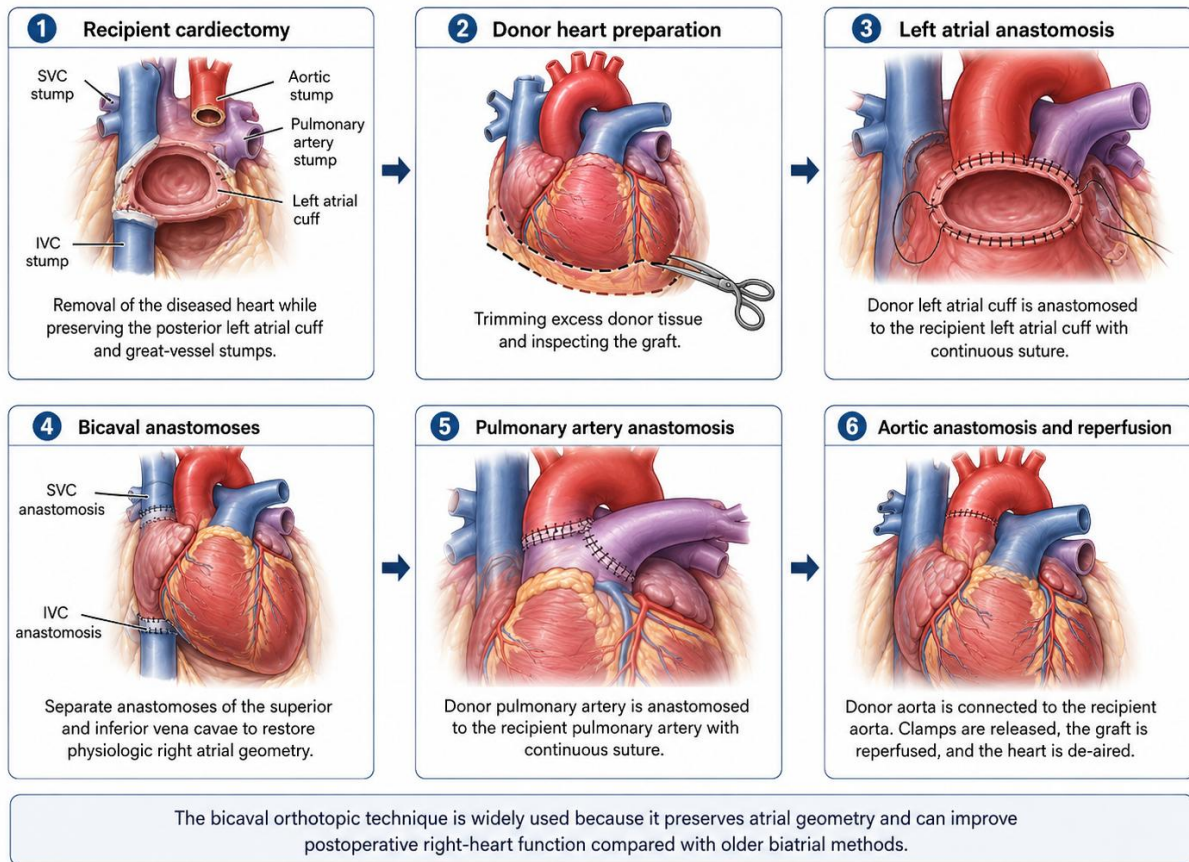


Figure 5. Orthotopic Heart Transplantation: Standard Bicaval Surgical Steps

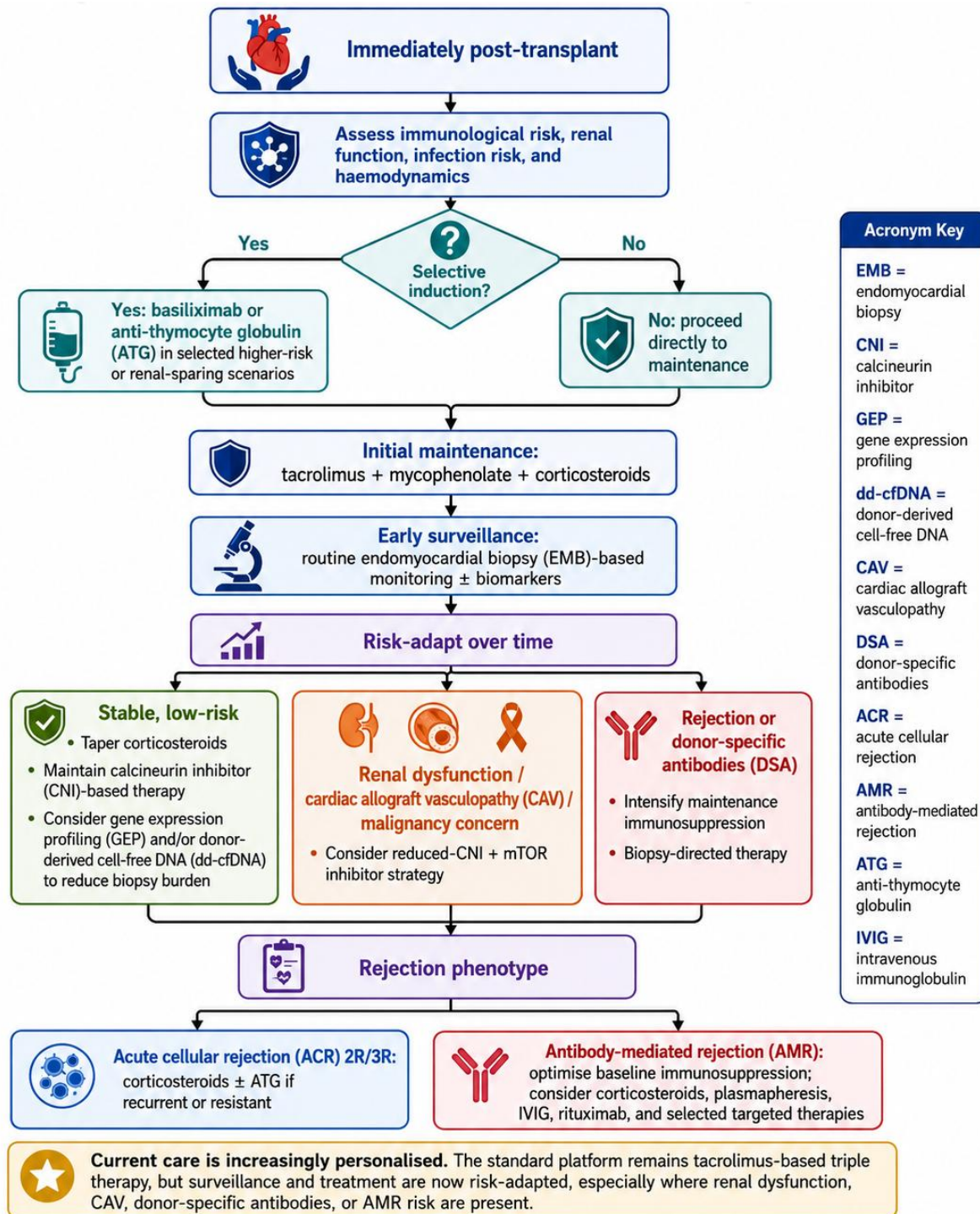


Figure 6. Described algorithm for immunosuppression and rejection surveillance

Current care is increasingly personalised. The standard platform remains tacrolimus-based triple therapy, but surveillance and treatment are now risk-adapted, especially where renal dysfunction, CAV, donor-specific antibodies or AMR risk are present. [24]

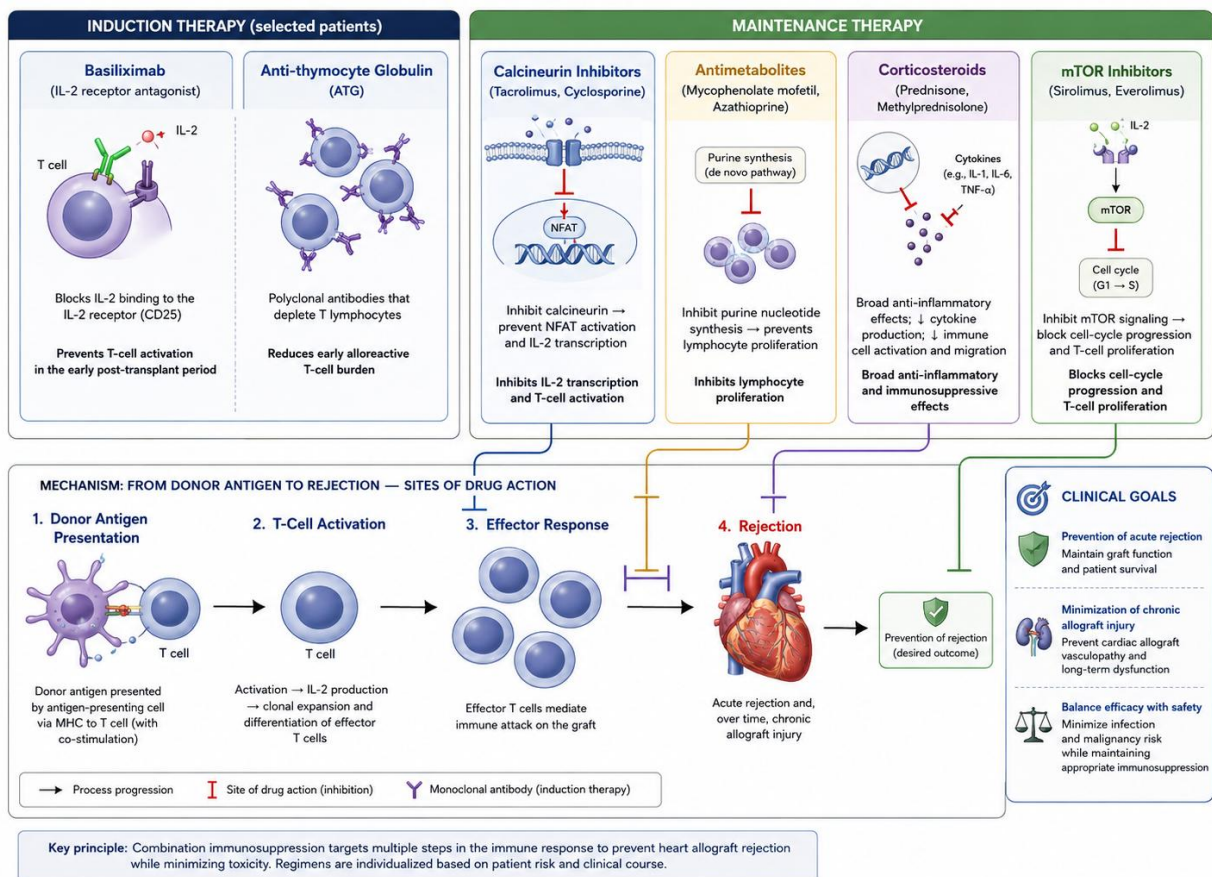


Figure 7. Immunosuppression in Heart Transplantation

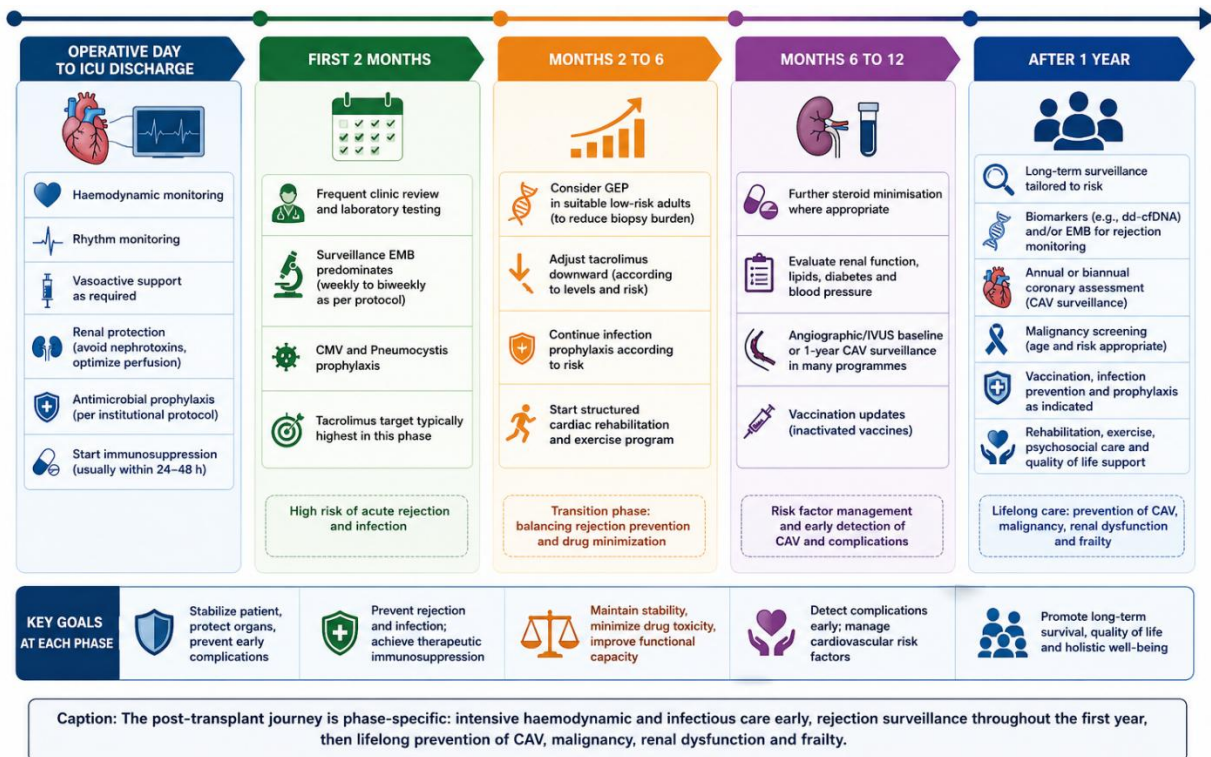


Figure 8. Described timeline of post-transplant care

The post-transplant journey is phase-specific: intensive haemodynamic and infectious care early, rejection surveillance throughout the first year, then lifelong prevention of CAV, malignancy, renal dysfunction and frailty. [25]

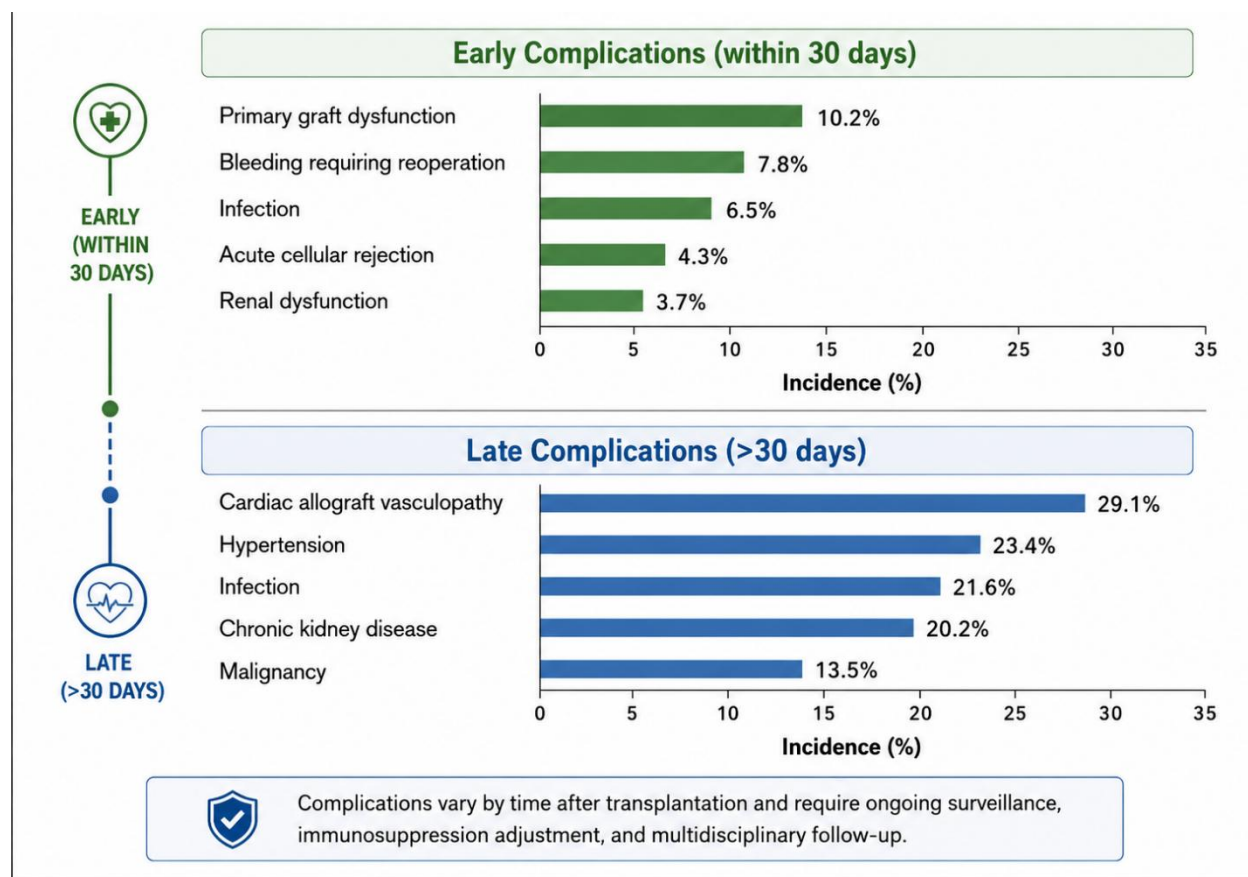


Figure 9. Early and Late Complications After Heart Transplantation

Conclusion, Limitations and Declarations

Conclusion: Heart transplantation remains one of the most successful and complex therapies in modern medicine. Current evidence supports a model of care that is more selective in candidate optimisation, more flexible in donor utilisation, more precise in immunological surveillance, and more attentive to long-term survivorship than at any previous point in the field’s history. The strongest contemporary gains are being made through earlier referral, multidisciplinary recipient assessment, better donor-recipient matching, DCD expansion, selective renal-sparing and CAV-oriented immunosuppression, and non-invasive augmentation of rejection monitoring. Even so, durable success is still limited by late complications—especially CAV, kidney disease, infection and malignancy—and by the ethical challenge of distributing scarce hearts fairly across populations and jurisdictions. [26]

Limitations: This manuscript is a structured narrative review rather than a formal systematic review or meta-analysis. Although primary studies, major guidelines and official registries were prioritised, some areas of practice remain supported by observational evidence, centre experience, or consensus rather than randomised trials.

Survival estimates are reported from heterogeneous registries and eras and should therefore be interpreted as ranges rather than directly pooled benchmarks. The manuscript also focuses primarily on adult practice, with paediatric data incorporated where they were especially instructive or where recent official reports were available.

Declarations

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